SANSKAR SINHA

Software Development Engineer (Geospatial Systems • Backend • Cloud Infrastructure)

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EDUCATION

KIET Group of Institutions, Ghaziabad — *B.Tech in Computer Science & Information Technology (GPA: 7.71/10.00)* [2024 – 2028]

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL (Spatial SQL), Go, C++
Geospatial & GIS Core: PostGIS, Shapely 2.x, PyProj, GeoPandas, GDAL/OGR, pyshp, defusedxml, R-Tree Spatial Indexing, Dynamic UTM/UPS Projections, WGS-84 Ellipsoidal Geodesics, MapLibre GL, Leaflet, GeoTIFF, COG
Backend & APIs: FastAPI, AsyncIO, Node.js, Express.js, RESTful APIs, WebSockets, Redis (Pub/Sub & Caching), BullMQ
Databases & ORM: PostgreSQL + PostGIS (ST_Area, ST_Distance, ST_Transform, ST_Intersects, GiST indexes), SQLite, SQLAlchemy Async, Prisma
Cloud & DevOps: Docker, Kubernetes, AWS (EC2, RDS PostgreSQL, S3, EKS, ECS, VPC, CloudWatch), CI/CD (GitHub Actions), Terraform, Linux/Bash
Testing & Security: Pytest (97% Coverage across 189 tests), Pytest-Cov, Ruff, OWASP API Security (XXE, Zip-Slip, CSV Injection defense)

WORK EXPERIENCE

ModelSuite.AI (Germany) — *SDE Intern (Remote)* **[Jul 2026 – Present]**
- Engineered high-throughput, asynchronous backend microservices using FastAPI, Node.js, PostgreSQL, Redis, and AWS.
- Containerized cloud workloads with multi-stage Docker builds; automated GitHub Actions CI/CD deployment pipelines.
- Automated cloud infrastructure provisioning via Terraform (IaC); configured system monitoring with Prometheus and Grafana.

SmarternX — *SDE Intern (Remote)* **[Jun 2026 – Jul 2026]**
- Built full-stack features and API endpoints using React.js, Node.js, and PostgreSQL for reliable cloud application infrastructure.
- Optimized relational schemas, database indexing, and automated deployment pipelines on AWS infrastructure.

SWOC (Script Winter of Code) — *Open Source Mentor (Remote)* **[Jan 2026 – Apr 2026]**
- Mentored 15+ open-source contributors on Git workflows, architectural design reviews, API debugging, and reliability best practices.

PROJECTS

Geospatial File Measurement & Intelligence API | *FastAPI, PostGIS, Shapely 2.x, PyProj, SQLAlchemy Async* — github.com/Sanskar225/geospatial-measurement-api
- Engineered an enterprise geospatial vector engine for Shapefile (.zip), KML 2.2, and KMZ ingestion, calculating metric area (m², ha, km²) and length (m, km) with zero native GDAL compilation overhead.
- Architected dynamic CRS reprojection to optimal local UTM zones (EPSG:326XX/327XX), cross-validating with WGS-84 ellipsoidal geodesics (pyproj.Geod) with <0.4% distortion variance across global mining coordinates (Pilbara, Chuquicamata, Dubai, Bengaluru).
- Built PostGIS-ready async persistence layer with streamed RFC 7946 GeoJSON/CSV exports and defense-in-depth hardening against Zip-Slip, zip-bombs, XXE (defusedxml), and formula injection.
- Achieved **97% automated test coverage across 189 tests** (Ruff-clean) and built a dark-mode Leaflet satellite viewer at /.

Drone Survey Spatial Analytics & Cadastral Ingestion Engine | *Python, PostGIS, GeoPandas, PostgreSQL, Docker*
- Designed a PostGIS spatial database engine using GiST indexing for low-latency queries (ST_Intersects, ST_Contains, ST_DWithin) for drone survey boundary verification.
- Implemented spatial SQL pipelines using ST_Transform to automate cadastral boundary reconciliation and plan-vs-actual survey overlay analysis.

ForgeStack – Architecture-Aware Cloud Infrastructure Generator | *Node.js, TypeScript, Docker, K8s, AWS, Terraform*
- Built a deterministic CLI (npx @sanskar225/create-forgestack) generating production Kubernetes, Terraform, and CI/CD configurations with graph-based dependency resolution supporting AWS ECS/EKS, RDS, and ElastiCache.

CERTIFICATIONS & ACHIEVEMENTS

- **AWS Certified Solutions Architect – Associate (2026)** | **AWS Certified Cloud Practitioner (2026)**
- **OpenCode'25 – Rank 65 / 2000+ (Top 3.2%):** Merged high-impact pull requests in open-source production repositories.
- **SIH 2025 University Winner & ECWOC Contributor:** Contributed to Odoo & HP Power Lab enterprise workflows.
- **300+ LeetCode Problems:** Strong foundation in Data Structures, Algorithms, REST APIs, and System Design.